From RANS Prediction to Hydrofoil Optimization: A Benchmark of Neural Surrogates

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Abstract

Neural flow surrogates are usually compared by field error, although their practical value often lies in the designs they select. We compare CNN-U-Net, a physics-regularized point network, FNO, and DeepONet on 381 OpenFOAM simulations of hydrofoils in water, using 64 cases from two unseen NACA families for validation. The models predict RANS fields, lift, drag, and pressure-based cavitation-inception risk, and then drive the same four-parameter hydrofoil shape-optimization loop. The physics-regularized network gives the best pressure accuracy ($R^2 = 0.966$), cavitation F1 score (0.921), and inference speed; DeepONet gives the most accurate force coefficients. Inference is 917–1801 times faster than OpenFOAM. Fresh OpenFOAM simulations confirm L/D gains of 2.3–79.7% for the selected designs, while a matched direct-CFD search reaches $L/D = 29.89$. Three surrogates select CFD-verified designs above that local-search reference, with the best reaching 45.96. Field accuracy, force accuracy, and optimization performance are related but not interchangeable.

1 Introduction

For a hydrofoil operating in water, shape design is an optimization problem wrapped around a flow solver. Each change in camber, thickness, or incidence requires another RANS solution, so even a modest search can consume hundreds of CFD evaluations; neural surrogates move most of that search off the solver by learning the map from geometry and operating conditions to flow and forces [Brunton et al., 2020]. The difficult question is not simply which network minimizes field error, but which one preserves the quantities that determine a useful design.

CNN-U-Nets emphasize local multiscale structure [Ronneberger et al., 2015]; FNO mixes information globally in Fourier space [Li et al., 2021, Kovachki et al., 2023]; DeepONet separates parameter and coordinate representations [Lu et al., 2021]; and physics-informed models constrain learning with differential equations [Raissi et al., 2019]. Their relative merits may therefore change as the target moves from pressure fields to forces and finally to shape optimization.

Airfoil-surrogate research has progressed from field prediction [Guo et al., 2016, Bhatnagar et al., 2019, Sekar et al., 2019, Thuerey et al., 2020] to benchmarks that evaluate surface forces, out-of-distribution accuracy, and computational cost [Bonnet et al., 2022, Li et al., 2023, Yagoubi et al., 2025]. Those metrics establish whether a model approximates CFD, but they do not by themselves establish whether it will make the same design decision. A small, structured force error can be amplified when an optimizer repeatedly queries a narrow region of design space. The missing experiment is therefore a controlled comparison that follows several surrogate families through the same optimizer and then checks the selected geometries with new CFD solutions. High-fidelity endpoint verification is standard in surrogate shape optimization, while a matched direct-CFD search provides a stronger reference when its cost is manageable [Li et al., 2022, Shukla et al., 2023].

That experiment is the contribution of this paper; we do not propose a new architecture. We train four representative model families on the same 381 RANS simulations and reserve two complete NACA families for validation. We compare fields, total forces, pressure-threshold cavitation screening, and runtime, then give every model the same three-start hydrodynamic shape optimization (HSO) problem. Fresh OpenFOAM solutions check the four surrogate winners, while a direct OpenFOAM search uses the same starts and budget as a reference. Cavitation risk here means inception inferred from C_p, not a multiphase prediction of cavity evolution.

2 Data and CFD Reference

2.1 Design space

The dataset contains NACA 0006, 0008, 0010, 0012, 0015, 0018, 2412, 2415, 4412, 4415, 4418, and 4421 profiles generated from the standard four-digit parameterization Abbott and von Doenhoff [1959]. For each profile, we simulate angles of attack $\alpha \in \{-4, -2, 0, 2, 4, 6, 8, 10\}^\circ$ and Reynolds numbers $Re \in \{10^5, 2 \times 10^5, 5 \times 10^5, 10^6\}$. All cases use liquid water with density 997 kg/m^3 and kinematic viscosity 10^{-6} m^2/s. With a 1 m chord, the freestream speed is set by $U_\infty = Re\nu/c$; the far-field absolute pressure and vapor pressure are 101325 and 2300 Pa.

Each case is solved with OpenFOAM `simpleFoam` [Weller et al., 1998]. For steady incompressible flow, the mean velocity $\bar{u}_i$ and pressure $\bar{p}$ satisfy

$$\frac{\partial \bar{u}_i}{\partial x_i} = 0, \tag{1}$$

$$\rho \bar{u}_j \frac{\partial \bar{u}_i}{\partial x_j} = -\frac{\partial \bar{p}}{\partial x_i} + \mu \frac{\partial^2 \bar{u}_i}{\partial x_j \partial x_j} - \rho \frac{\partial \overline{u'_i u'_j}}{\partial x_j}. \tag{2}$$

The Reynolds stress is closed with the k–ω SST model [Menter, 1994]. Angled freestream velocity is imposed at the inlet and far boundary, and OpenFOAM's `forceCoeffs` supplies total lift and drag after 300 iterations. Cell-centered fields are interpolated to a common 128×64 Cartesian grid over $x/c \in [-1, 2]$ and $y/c \in [-0.75, 0.75]$. Inputs include coordinates, operating parameters, three NACA descriptors, a fluid mask, and signed distance to the profile.

Three simulations with $|C_L|$ or $|C_D| > 5$ are discarded before splitting, leaving 381 cases. All 64 cases from NACA 0015 and NACA 2412 form the validation set; the remaining 317 cases train the models. Because complete families are reserved, neither validation geometry appears in training at a different operating condition. All retained solves reach iteration 300. The 95th-percentile maximum initial residual at that iteration is 1.27×10^{-4}; the corresponding late-window spans are 0.0059 in C_L and 2.65×10^{-4} in C_D.

2.2 Targets and cavitation-risk definition

The common output vector contains U_x, U_y, absolute pressure p, eddy viscosity, k, ω, C_p, cavitation margin, C_L, and C_D. Grid fields use masked normalized mean-squared error, while the case-level force targets use normalized mean-squared error. OpenFOAM's kinematic gauge pressure p_k is converted to absolute pressure as

$$p = p_\infty + \rho p_k, \qquad C_p = \frac{p - p_\infty}{\frac{1}{2}\rho U_\infty^2}. \tag{3}$$

For an ambient-pressure value p_a, the pressure-based inception margin is

$$M(x, y; p_a) = p_a + C_p(x, y)\tfrac{1}{2}\rho U_\infty^2 - p_v. \tag{4}$$

A cell is labeled at risk when $M < 0$. We evaluate this label at $p_a \in \{2500, 3000, 5000, 10000, 25000, 101325\}$ Pa. This sweep creates a useful classification test from the predicted C_p field without representing multiphase physics.

3 Surrogates and Optimization

3.1 Model families

The CNN-U-Net has three encoder/decoder levels, skip connections, group normalization, GELU activations, and base width 16. The FNO has width 32, four operator blocks, and 12 retained Fourier modes per direction Ronneberger et al. [2015], Li et al. [2021]:

$$h_{l+1} = \sigma\big(W_l h_l + \mathcal{F}^{-1}(R_l \mathcal{F}(h_l))\big) . \tag{5}$$

The DeepONet uses three-layer, width-64 branch and trunk networks with 64 basis functions. Its output has the form

$$\widehat{y}_k(\mathbf{x}; \boldsymbol{\mu}) = \mathbf{b}_k(\boldsymbol{\mu})^\top \mathbf{t}_k(\mathbf{x})/\sqrt{64} + c_k. \tag{6}$$

following Lu et al. [2021]. The physics-regularized point network has three tanh layers of width 64 and uses 1,024 sampled points per case. Its supervised loss is augmented by 10^{-4} times the continuity and first-order momentum residual loss Raissi et al. [2019]. This is a lightweight physics regularizer, not a data-free PINN or complete RANS residual.

All models use AdamW with learning rate 2×10^{-3}, weight decay 10^{-4}, at most 120 epochs, patience 20, gradient clipping, and seed 7. Normalization uses training cases only. The best checkpoint is selected by validation loss. Model size ranges from 9,930 to 595,914 parameters (Table 1).

Optimization uses dedicated C_L and C_D output heads supervised by OpenFOAM's total force coefficients. Pressure integration on the coarse common grid is not used for the objective because it omits viscous traction and under-resolves the wall.

3.2 Common optimization protocol

At $Re = 5 \times 10^5$, each surrogate drives the same HSO loop. The design vector $\theta = (m, p, t, \alpha)$ contains maximum camber, camber location, thickness, and incidence, with bounds $m \in [0, 0.09]$, $p \in [0.1, 0.9]$, $t \in [0.06, 0.20]$, and $\alpha \in [-4, 10]^\circ$. We launch Nelder–Mead [Nelder and Mead, 1965] from NACA 0012, 2412, and 4415 and give every run the same 40-iteration budget. The objective is

$$J(\theta) = \frac{C_L}{|C_D| + 10^{-6}} - 10P_M - 5f_M, \tag{7}$$

where P_M penalizes a predicted minimum cavitation margin below 5000 Pa and f_M is the fraction of cells below that margin. The search calls only the surrogate; OpenFOAM is not used inside the loop. Since the nominal-pressure margin stays above the threshold for every selected design, the penalty is inactive in this experiment. We return the best design observed within the fixed budget, select the strongest of the three starts for each architecture, and then generate a new mesh and independent OpenFOAM simulation for that winner.

The direct-CFD baseline repeats this protocol with the same starts, bounds, objective, tolerances, and 40-iteration budget, but OpenFOAM supplies every objective value. It is distinct from the one-solve endpoint checks of the surrogate winners.

4 Results

4.1 Validation accuracy and computational cost

Table 1 reports results on the 64-case validation set. The physics-regularized network is best for pressure and C_p, with pressure RMSE 6.78 Pa and $R^2 = 0.966$. DeepONet gives the best U_x, U_y, C_L, and C_D coefficients. The grid operators are weaker in this low-data

Table 1: Validation performance, model size, and cost. Force RMSE is computed case-wise against OpenFOAM total coefficients.

Model	p RMSE	p R^2	C_L RMSE	C_D RMSE	Params.	Speedup
CNN-U-Net	28.52	0.392	0.0991	0.00453	485,178	1,148×
Physics-reg. MLP	6.78	0.966	0.0245	0.00204	9,930	1,801×
FNO	22.73	0.614	0.1383	0.00957	595,914	1,028×
DeepONet	8.55	0.945	0.0268	0.00184	100,746	917×

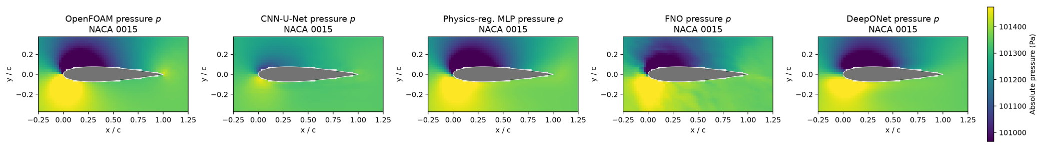

Figure 1: Absolute-pressure prediction for a representative NACA 0015 validation case (case 160), shown over $x/c \in [-0.25, 1.25]$ and $y/c \in [-0.375, 0.375]$. Every panel uses the same color scale.

setting. FNO retains useful force trends, while CNN-U-Net predicts C_p and forces better than absolute pressure or streamwise velocity.

Measured training times are 868, 104, 547, and 54 s for CNN-U-Net, the physics-regularized network, FNO, and DeepONet, respectively. Median OpenFOAM execution time is 12.4 s per case, while pressure-field inference requires 0.0069–0.0135 s, yielding 917–1801× speedups. The benchmark used an 8-core Apple M2 Mac mini with 16 GB memory; PyTorch 2.13 ran on CPU and single-process OpenFOAM 9 ran in Docker. These numbers measure repeated local inference and exclude data generation and mesh construction.

Figure 1 shows a representative validation case from the unseen NACA 0015 family. The pointwise models reproduce the leading-edge pressure structure most closely; CNN-U-Net smooths the extrema and FNO introduces spatial texture. A common color scale makes those differences directly comparable across all five panels.

4.2 Cavitation-inception screening

The pressure-threshold labels are highly imbalanced, so accuracy is not useful here. The physics-regularized network achieves precision 0.927, recall 0.916, and F1 0.921. DeepONet follows at F1 0.913. CNN-U-Net has the highest recall (0.958) but lower precision (0.796), while FNO obtains F1 0.771. Twenty-four validation case/ambient-pressure combinations contain at-risk cells. This measures pressure-based inception screening, not cavity dynamics. Multiphase labels are required to predict vapor fraction, cavity length, or shedding Brennen [1995].

4.3 Shape optimization against direct CFD

Figure 2 reports the complete L/D comparison: the OpenFOAM value of the starting foil, the surrogate estimate at its selected design, and the result of remeshing that design and solving it once in OpenFOAM. This one-solve endpoint check is distinct from the direct baseline, which places OpenFOAM inside the matched optimizer.

All four surrogate winners improve on their starts. The physics-regularized network moves from $L/D = 25.57$ to 45.96, an absolute gain of 20.39 (79.7%); DeepONet gains 14.97, CNN-U-Net gains 12.55, and FNO gains 0.58. Direct CFD reaches 17.80, 27.33, and 29.89 from NACA 0012, 2412, and 4415, respectively, using 266 evaluations in total. Its best gain is 4.32 (16.9%). The selected (m, p, t, α) values are $(0.0325, 0.550, 0.060, 3.06°)$, $(0.0667, 0.388, 0.060, 1.24°)$, $(0.0391, 0.372, 0.168, 4.23°)$, and $(0.0335, 0.328, 0.060, 1.60°)$ for

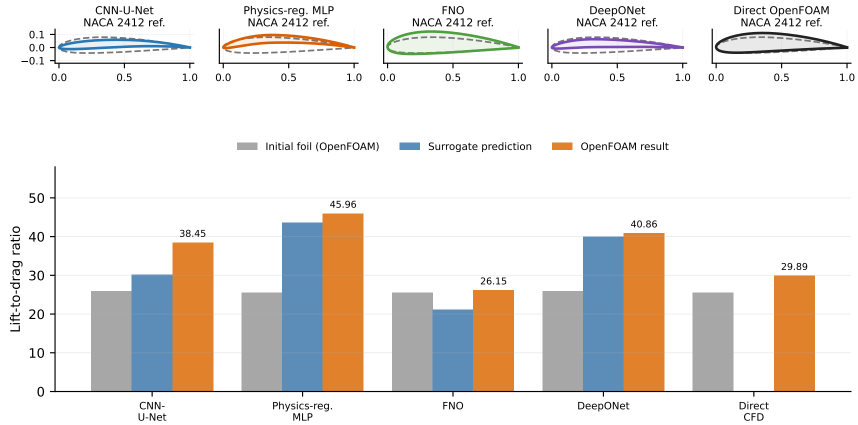

Figure 2: Top: selected shapes against the same dashed NACA 2412 geometric reference. The reference is for visual comparison, not necessarily the optimizer initializer. Bottom: initial-foil OpenFOAM values, surrogate predictions, fresh OpenFOAM endpoint checks, and direct CFD.

CNN-U-Net, the physics-regularized network, FNO, and DeepONet, respectively; direct CFD selects $(0.0407, 0.417, 0.144, 3.98°)$.

5 Discussion

Accuracy and decision quality. The results expose two distinct error modes. DeepONet has the lowest endpoint L/D error (2.1%), matching its strong force metrics, yet the physics-regularized network selects the better design: $L/D = 45.96$ rather than 40.86. CNN-U-Net and FNO underpredict by 21.5% and 19.3%, but only CNN-U-Net locates a large gain. Value calibration and candidate ranking are therefore separate capabilities; endpoint CFD changes the model comparison rather than merely confirming the force-error table.

The matched direct search reaches only $L/D = 29.89$. CNN-U-Net, DeepONet, and the physics-regularized model exceed it by 8.56, 10.97, and 16.07 after CFD validation, while FNO remains below it. The direct run is not a global optimum; it shows that learned objectives can lead the same budgeted local optimizer to different basins.

Search behavior. Initialization matters. The physics-regularized network reaches predicted $L/D = 43.55$ and 43.62 from the two cambered starts but only 22.35 from NACA 0012; FNO stays between 16.18 and 21.11. The three stronger designs approach 6% thickness, whereas FNO selects 16.8%. The pointwise models thus learn a sharper preference for thin, cambered profiles in this region.

Cost and cavitation. Across the three starts, the models require 199–249 evaluations, approximately 41–52 minutes if replaced one-for-one by the measured CFD solve versus 1.4–3.0 s of surrogate inference. The measured direct baseline uses 266 CFD evaluations and 38.5 minutes of summed wall time; each surrogate additionally requires one endpoint solve.

All selected designs have a nominal predicted cavitation margin near 99 kPa, so the penalty in Eq. (7) is inactive. The cavitation results establish screening accuracy, not a cavitation-shaped optimum; demonstrating that tradeoff requires reduced ambient pressure or multiphase labels. The remaining scope is one seed, steady 2D RANS, and the NACA four-digit family; grid convergence and external force validation remain future checks.

6 Conclusion

We followed four neural surrogates through the complete path from RANS-field prediction to a design decision. On unseen NACA families, the physics-regularized point network gives the best pressure result ($R^2 = 0.966$) and cavitation-inception F1 score (0.921), while DeepONet gives the most accurate lift and drag. Every model evaluates a pressure field in 6.9–13.5 ms, corresponding to 917–1801× speedup over the measured OpenFOAM runs.

The HSO experiment changes the ranking from a simple accuracy table. DeepONet predicts its winner most accurately, but the physics-regularized network finds the best design, increasing OpenFOAM L/D from 25.57 to 45.96. CNN-U-Net and DeepONet also find substantial gains, from 25.89 to 38.45 and 40.86, respectively, whereas FNO reaches only 26.15 from 25.57. The direct-CFD search reaches 29.89, confirming that evaluator choice changes which basin a limited local search reaches. A useful scientific surrogate should therefore be judged on both its validation predictions and the consequence of acting on them.

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